Jingxuan Ma

+1 (410) 240-4366 | jma107@jh.edu | LinkedIn | Portfolio

Data Science M.S. student at Johns Hopkins with a background in Applied Statistics and data analytics. Skilled in R, advanced SQL, Python ETL pipelines using Apache Airflow, Tableau dashboard development, operations automation, stakeholder collaboration, and AI prompt engineering to accelerate data analysis.

EDUCATION

Johns Hopkins University August 2025 - June 2027
Master of Science in Data Science Baltimore, US

- **Coursework:** Database Systems | Introduction to Algorithms | Nonlinear Optimization |Human-Computer Interaction

University College Cork September 2021 - June 2025
Bachelor of Science in Actuarial Sciences Cork, Ireland | Beijing, China

- **Coursework:** Probability & Statistics | Actuarial Mathematics | Financial Mathematics | Regression Analysis | Statistical Computing

PROFESSIONAL EXPERIENCE

AI Agent Intern | Beijing Yiling Network Technology Co., Ltd. **Beijing, China | June 2026 - Present**

- Automated end-to-end job-application prep: JD parsing, inventory projection, DOCX injection, and one-page PDF validation to reduce manual resume formatting work
- Designed format-lock and quality-gate checks (shell fingerprint, hyperlink preservation, Word PDF one-page validation, evidence-linked bullets) so delivery copies keep master fonts, margins, and list styles without rebuilding the document
- Documented failure modes (multi-page spill, broken hyperlinks, stacked summaries) and encoded them as blocking quality gates in the agent pipeline

Data Analyst Intern | Shenwan Hongyuan Securities Co., Ltd. Beijing, China | June 2024 - August 2024

- Faced with the need to evaluate whether a pricing model library could meet both runtime and maintainability requirements, conducted a structured feasibility analysis across Python, pure & optimized C++, Eigen vectorization, OpenMP, and ctypes-based Python/C++ integration
- Built and benchmarked a 100,000-path Monte Carlo pricing simulation, applying compiler optimization, vectorized matrix operations, and multithreaded random-path generation to isolate major bottlenecks in computation-intensive pricing workflows
- Delivered a hybrid architecture recommendation that assigned heavy simulation and statistical computation to C++ while keeping configuration, preprocessing, and visualization in Python; reduced runtime from approx. 33.4s to approx. 1.4s with OpenMP optimization

Data Analyst Intern | Yinhua Fund Management Co., Ltd. Beijing, China | June 2023 - August 2023

- Given the need to support campaign review and market reporting, collected, organized, and cleaned market and customer data from multiple business materials, creating structured datasets for trend analysis and management communication
- Used Python, Excel, and visualization techniques to analyze market patterns, prepare charts and analytical materials, and convert raw performance data into insights that could support investment-product marketing decisions
- Prepared concise reporting assets for internal stakeholders by translating quantitative findings into business narratives, improving clarity around customer behavior, campaign performance, and market trends

PROJECTS

Credit Risk Prediction Model | Python, SQL, scikit-learn, XGBoost, R Independent Project

- To build and adapt algorithms for complex risk use cases, designed an end-to-end predictive pipeline integrating SQL-style extraction, missing-value treatment, feature engineering, and statistical modeling to estimate expected claim costs and credit default behavior
- Applied advanced statistics and machine learning libraries (scikit-learn, XGBoost) to train, evaluate, and benchmark regression and tree-based models, leveraging ROC-AUC, F1-score, and cost drivers to balance predictive accuracy with business interpretability.
- Extended the framework with stochastic modeling via Monte Carlo simulations to analyze skewed loss distributions, interpreting error patterns to translate quantitative outputs into risk-monitoring thresholds and optimized decision-making insights.

Tesla Vehicle Quality & Risk Analytics Pipeline | Python, Apache Airflow, SQL, Tableau, NHTSA API Independent Project

- Built an end-to-end ETL pipeline extracting 11,349 NHTSA complaints and 60 recall actions across Tesla Models 3/S/X/Y (2015–2024); engineered dynamic model-name discovery and ID-based deduplication to reduce 30,777 raw records to 11,349 verified distinct complaints
- Engineered frequency- and severity-based risk indicators (crash/fire/injury-weighted scores by model-year and component) in SQL, applying an actuarial-style frequency × severity loss framework to quantify and rank vehicle quality risk across product lines
- Built and published an interactive Tableau dashboard visualizing component-level risk rankings and recall-lag trends (first complaint to official recall), translating 11,349 complaint records into decision-ready risk insights for quality and safety stakeholders.

• COMPETITIONS

Mathematical Contest in Modeling | Team Leader Remote | February 2023

- Led a student modeling team through problem scoping, data cleaning, indicator screening, visualization, model development, and technical report writing, coordinating responsibilities to deliver a structured solution under time constraints

Mathematical Modeling for College Students | Team Member Remote | September 2022

- Collected and processed data for statistical modeling tasks; built models using Python and R, analyzed results with SPSS, and contributed to result interpretation and final written recommendations

SKILLS & CERTIFICATIONS

Python, R, SQL, stakeholder reporting, Tableau, Apache Airflow, data cleaning, feature engineering, exploratory analysis, Pandas, NumPy, scikit-learn, XGBoost, Monte Carlo Simulation, Credit Risk, Claims Modeling